

DV2627 Advanced ML

Lecture: Semi-supervised Learning

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Outline

- Introduction
- Examples
- Weak supervision - incomplete versus inexact
- How semi-supervised learning works
- Semi-supervised learning techniques
 - Inductive and transductive methods
 - Meta algorithms
 - Self-training and co-training
 - Specific algorithms
 - Transductive SVM, graph-based methods, DL, semi-supervised clustering (COP- k -means)
- Active learning
- Conclusions
- References

Introduction (1)

- Unsupervised versus Supervised learning
 - Two extreme learning paradigms
 - **Unsupervised learning**
 - collection of data objects without any labels
 - easy to collect
 - **Supervised learning**
 - each object tagged with a class
 - laborious job
- **Semi-supervised learning (SSL)**
 - Real life applications are somewhere in between

Introduction (2)

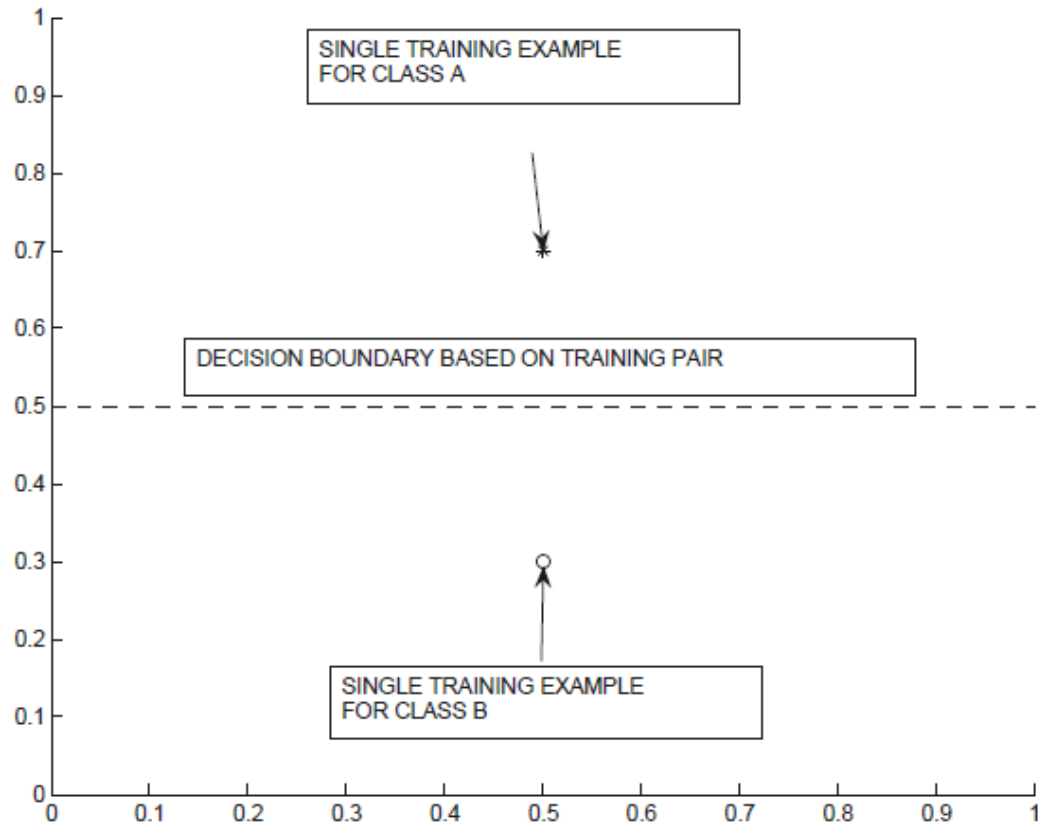
- (Fully) supervised learning (traditional classification)
- Partially (semi-) supervised learning
 - Learning with a small set of labeled examples and a large set of unlabeled examples
 - Learning with positive and unlabeled examples (no labeled negative examples)

Introduction (3)

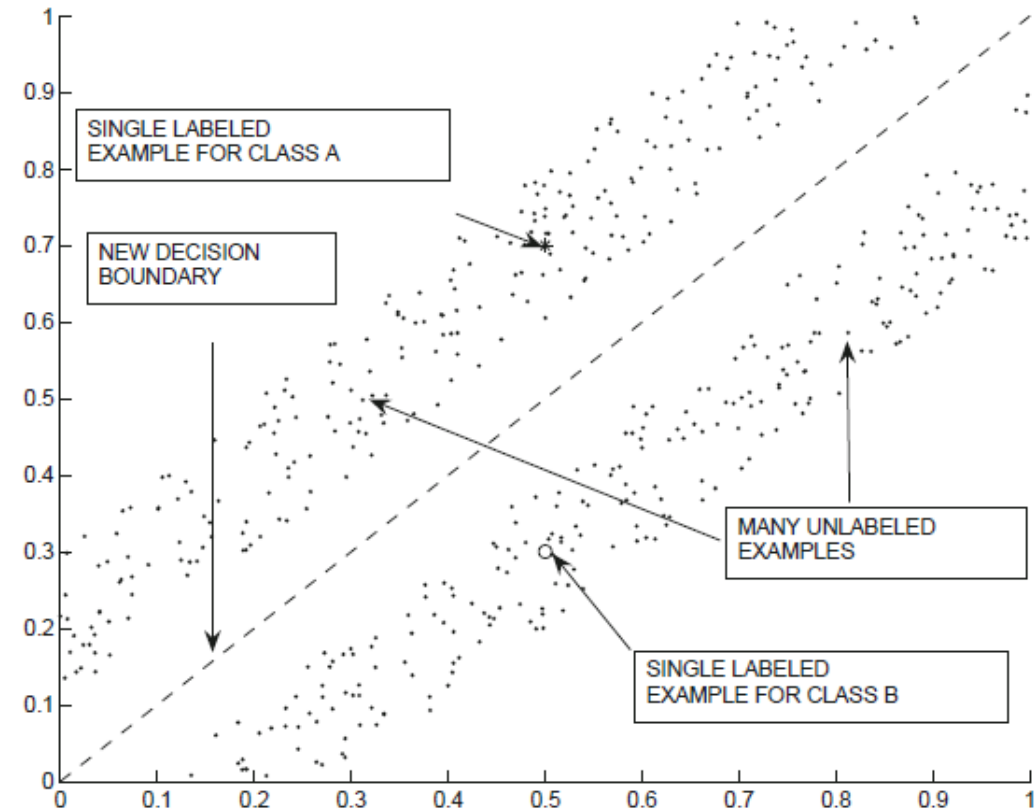
- Labeled data is expensive and hard to acquire
- Unlabeled data is often copiously available
- Unlabeled data is useful
 - Unlabeled data can be used to estimate the low-dimensional **manifold structure** of the data
 - Unlabeled data can be used to estimate the **joint probability distribution** of features

Example 1

Class variables are likely to **vary smoothly over dense regions**



(a) Labeled data



(b) Labeled and unlabeled data

Example 2

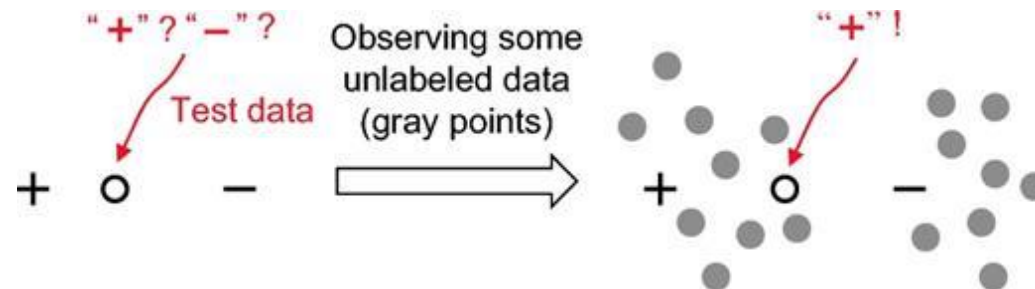
- The goal is to determine whether documents belong to the "Science" category.
- In **labeled** data, we found the word "Physics" is associated with the "Science" category
- In **unlabeled** data, we found the word "Einstein" often co-occur with "Physics"
- Thus, the **unlabeled** documents provide the insight that the word "Einstein" is also relevant to the "Science" category

Weak supervision - incomplete versus inexact

- **Incomplete supervision** is a (**semi-supervised**) learning scenario in which only a subset of training data are given with labels whereas the other data remain unlabeled
 - **Active learning** assumes that the ground-truth labels of unlabeled instances can be queried from an oracle (e.g., a human expert)
 - **Semi-supervised learning** attempts to exploit unlabeled data without querying human experts
- **Inexact supervision** is a (**multi-instance**) learning scenario in which only coarse-grained labels are given

How semi-supervised learning works

- Most **semi-supervised learning** approaches make strong model assumptions (guesses).
- Some commonly used **assumptions**:
 - The cluster data objects are from the same class
 - Data can be represented as a mixture of parameterized distributions
 - Decision boundaries should go through non-dense areas of the data
 - Model should be as simple as possible (Occam razor)



Semi-supervised Learning Techniques (1)

- Transductive and inductive methods
 - **Transductive methods**: only label the available unlabeled data – not generating a classifier
 - **Inductive methods**: not only produce labels for unlabeled data, but also generate a classifier
- Algorithmic methods
 - **Classifier-based methods**: start from an initial classifier, and iteratively enhance it
 - **Data-based methods**: find an inherent geometry in the data, and use the geometry to find a good classifier
- Meta-algorithms and specific algorithms

Semi-supervised Learning Techniques (2)

- **Meta-algorithms** can use any existing classification algorithm as a subroutine
 - **Self-Training**
 - **Co-training** (Multi-modal learning)
 - Assume that each modality is sufficient for classification independently
 - Uses multiple modal classifiers to label/move instances of unlabeled set U to labeled set L
 - **Ensemble learning**
 - Train multiple diverse models on L . Those instances in U which most models agree on are placed in L .

Self-Training

- The Procedure

- Initial labeled set L and unlabeled set U
 1. Use algorithm $\mathcal{C}$ trained on the current labeled set to identify the k instances in the unlabeled data U for which the classifier $\mathcal{C}$ is the **most confident**
 2. Assign labels to the k most confidently predicted instances and add them to L . Remove these instances from U
 3. Repeat 1 and 2

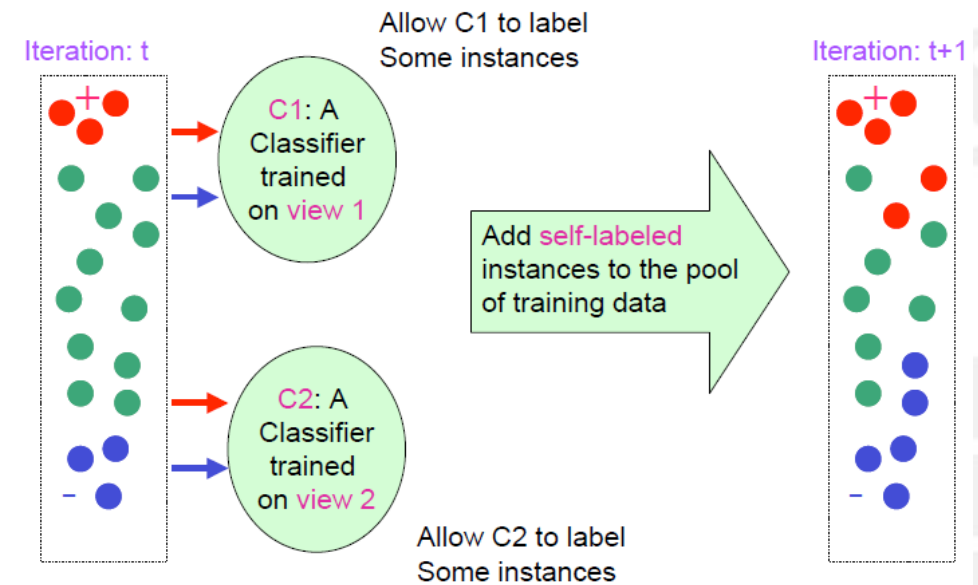
- Overfitting

- Addition to predicted labels may propagate errors

Co-Training (1)

- The Procedure

- Two disjoint feature groups: F_1 (view 1) and F_2 (view 2)
- Two labeled sets L_1 and L_2
 1. Train classifier $\mathcal{C}_1$ using labeled set L_1 and feature set F_1 , and add k most confidently predicted instances from unlabeled set U to training data set L_2 for classifier $\mathcal{C}_2$
 2. Train classifier $\mathcal{C}_2$ using labeled set L_2 and feature set F_2 , and add k most confidently predicted instances from unlabeled set U to training data set L_1 for classifier $\mathcal{C}_1$
 3. Repeat 1 and 2



Co-Training (2)

- Many problems have two different sources of info you can use to determine label
- For example, classifying webpages: can use **words on page** or **words on links pointing to the page**

Co-Training (2)

Idea: Use small labeled sample to learn initial rules, e.g.,

- “my advisor” pointing to a page is a good indicator it is a faculty home page.
- “teaching experience in” on a page is a good indicator it is a faculty home page.

Prof. Veselka Boeva My Advisor



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Professor of Computer Science

Veselka Boeva is Professor of Computer Science at Blekinge Institute of Technology, Sweden. She received an M. Sc. degree in Mathematics with specialization in Computer Science at the Plovdiv University, Bulgaria. She was awarded a Ph. D. degree in Computer Science from the Technical University of Sofia branch Plovdiv, Bulgaria in 2000. Veselka has a long teaching experience in Computer Science and research experience in the field of Artificial Intelligence, Data Mining, Machine Learning, Data Analysis and Bioinformatics. She was co-chair of the Bioinformatics Special Interest Group (BI SIG) for more than 10 years, an initiative founded at the Technical University-Plovdiv in 2004. The BI SIG was the first research group in the field of bioinformatics founded at that time in Bulgaria. Her research has been partly funded by successful research projects (e.g., a Bulgarian National Science Fund project and a KKS project) and a string of different research fellowships: a Post-Doc Research Fellowship supported by a Swedish Institute grant; a Post-Doc Research Fellowship supported by a NATO grant; a Post-Doc Research Fellowship supported by a Belgian Federal Science Policy Office grant and a Research Fellowship of German Academic Exchange Service (DAAD). As a result, she has visited and worked at various European universities in Slovakia, Belgium, Germany and Sweden. She is also very actively engaged as an external expert for evaluating research and higher education projects for FP7, H2020 and Horizon Europe programs of the EU.

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x_2 - Text info

Co-Training (2)

Idea: Use small labeled sample to learn initial rules, e.g.

- “my advisor” pointing to a page is a good indicator it is a faculty home page.
- “teaching experience in” on a page is a good indicator it is a faculty home page.
- Then look for **unlabeled** examples where one rule is confident and the other is not. Have it to label the example for the other.
- Training two classifiers, one on each type of info. Using each to help train the other.

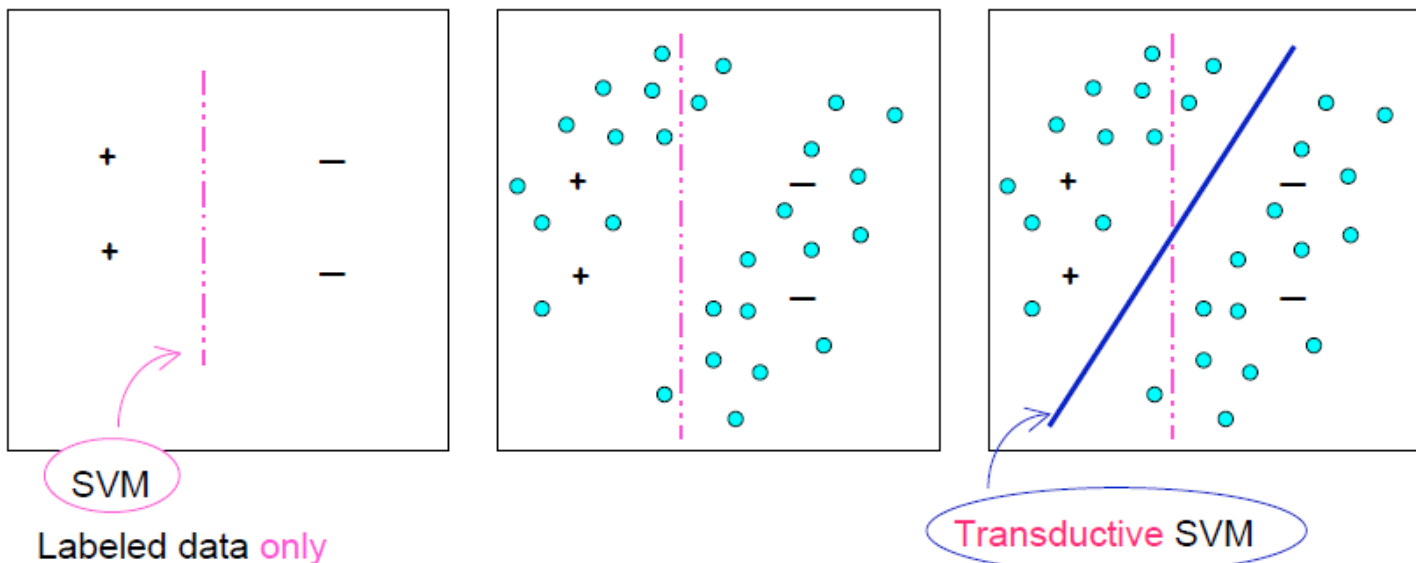
Semi-supervised Learning Techniques (3)

- **Specific Algorithms**

- Transductive support vector machines
- Graph-based semi-supervised learning
- DL semi-supervised learning
- Semi-supervised clustering

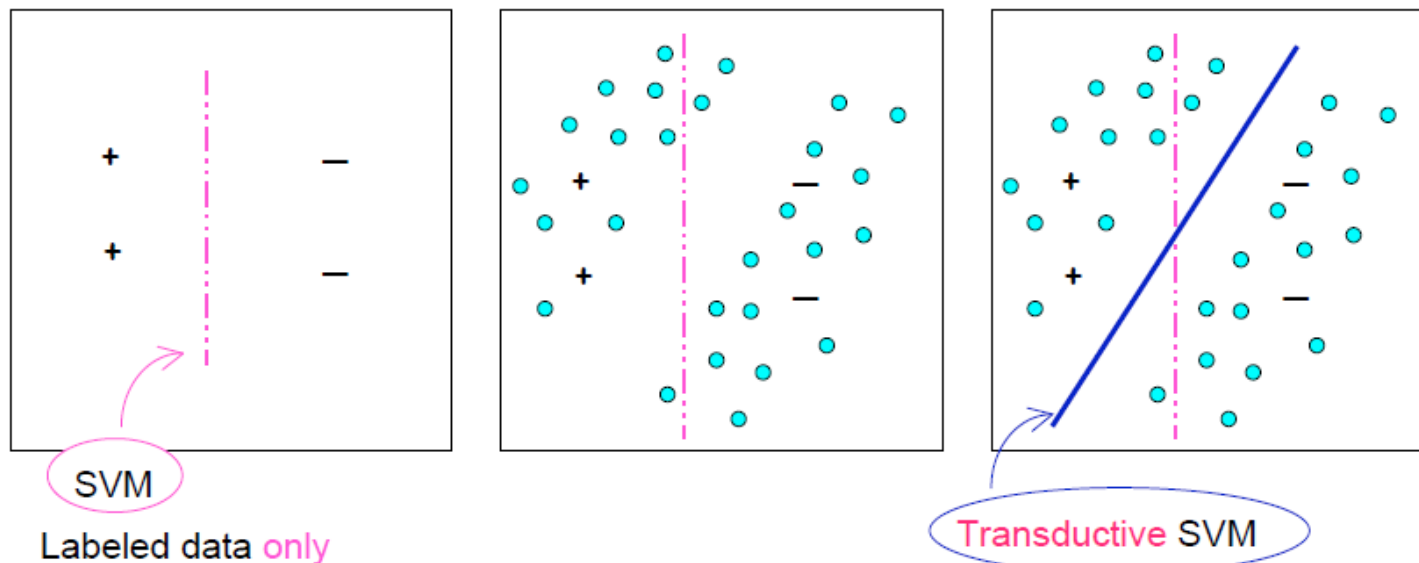
Transductive SVM (1)

- Maximize margin of both L and U . Decision surface placed in non-dense spaces
 - Assumes classes are "well-separated"
 - Can also try to simultaneously maintain class proportion on both sides similar to labeled proportion



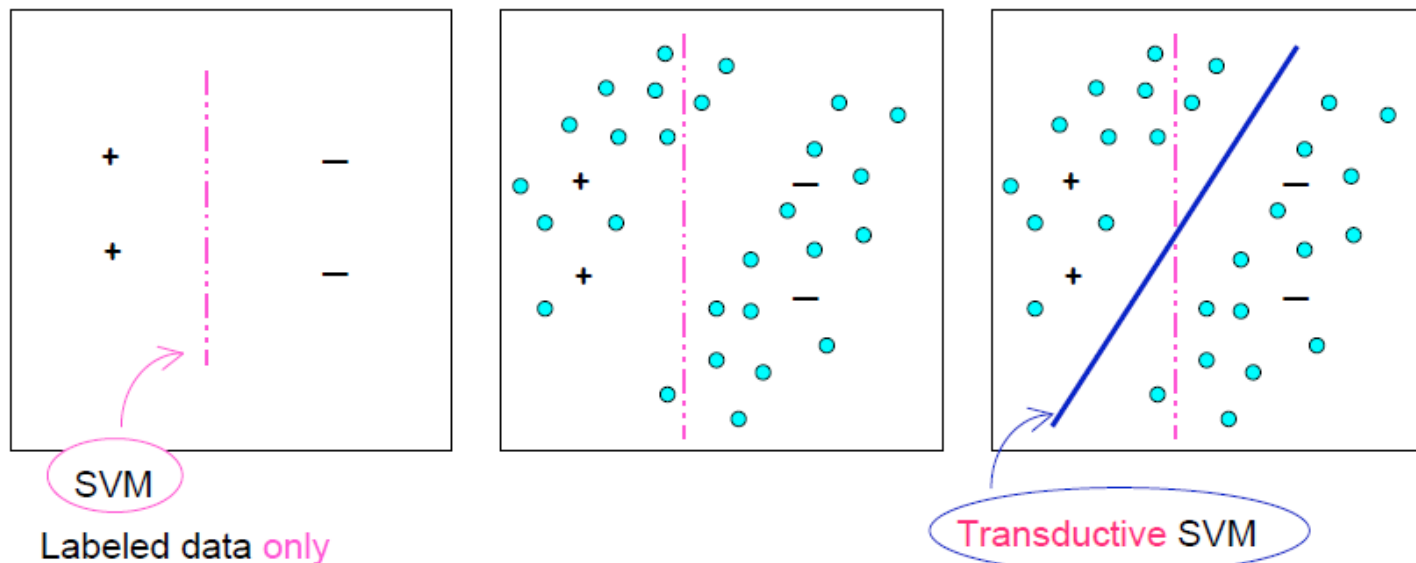
Transductive SVM (2)

- Unfortunately, optimization problem is NP-hard. The algorithm instead does **local optimization**
 - Start with large margin over labeled data. Induces labels on unlabeled data U
 - Then try flipping labels in greedy fashion



Transductive SVM (2)

- Unfortunately, optimization problem is NP-hard. The algorithm instead does **local optimization**
 - Start with large margin over labeled data. Induces labels on unlabeled data U
 - Then try flipping labels in greedy fashion
- Quite successful on text data, used also in cancer classification, image retrieval, ...

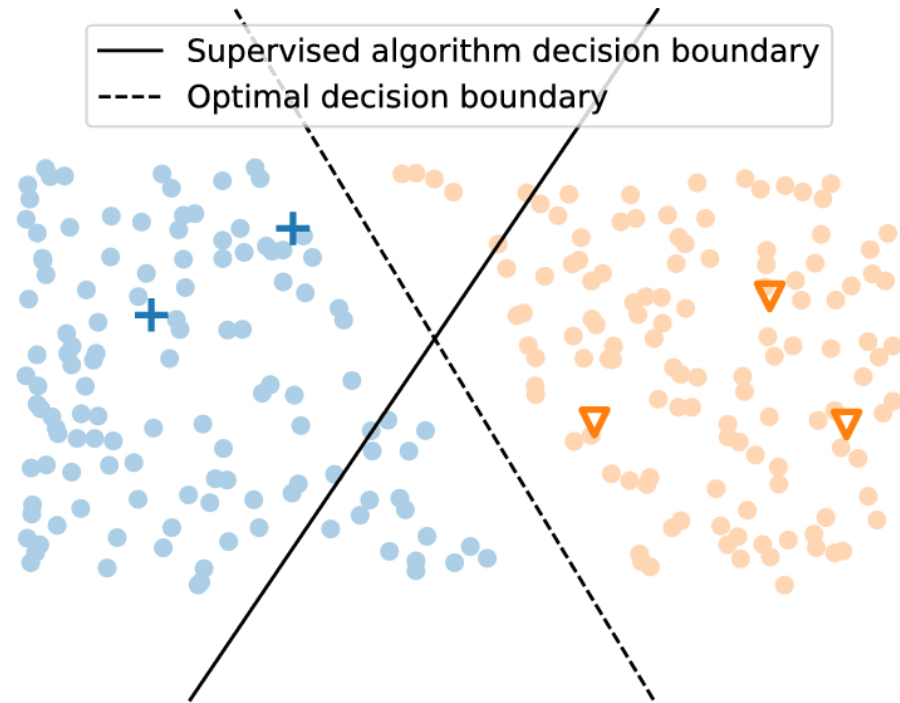


Assumptions of semi-supervised learning (1)

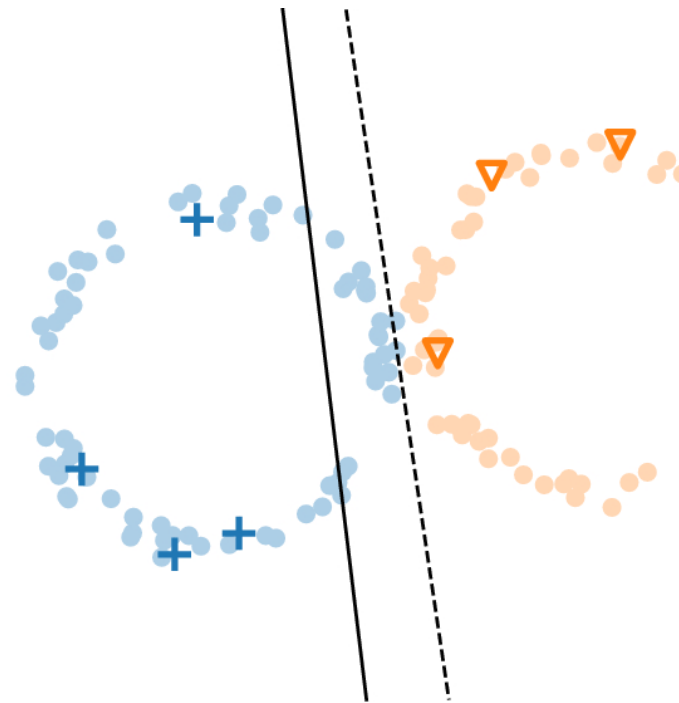
- **Semi-supervised learning assumptions** formalize the types of expected interaction between label and unlabeled data
 - smoothness assumption
 - if two samples x and x' are close in the input space, their labels y and y' should be the same
 - low-density assumption
 - the decision boundary should not pass through high-density areas in the input space
 - manifold assumption
 - data points on the same low-dimensional manifold should have the same label
 - cluster assumption
 - data points belonging to the same cluster belong to the same class
 - the cluster assumption is a generalization of the other assumptions

Assumptions of semi-supervised learning (2)

- **Illustration of semi-supervised learning assumptions**



(a) Smoothness and low-density assumptions.

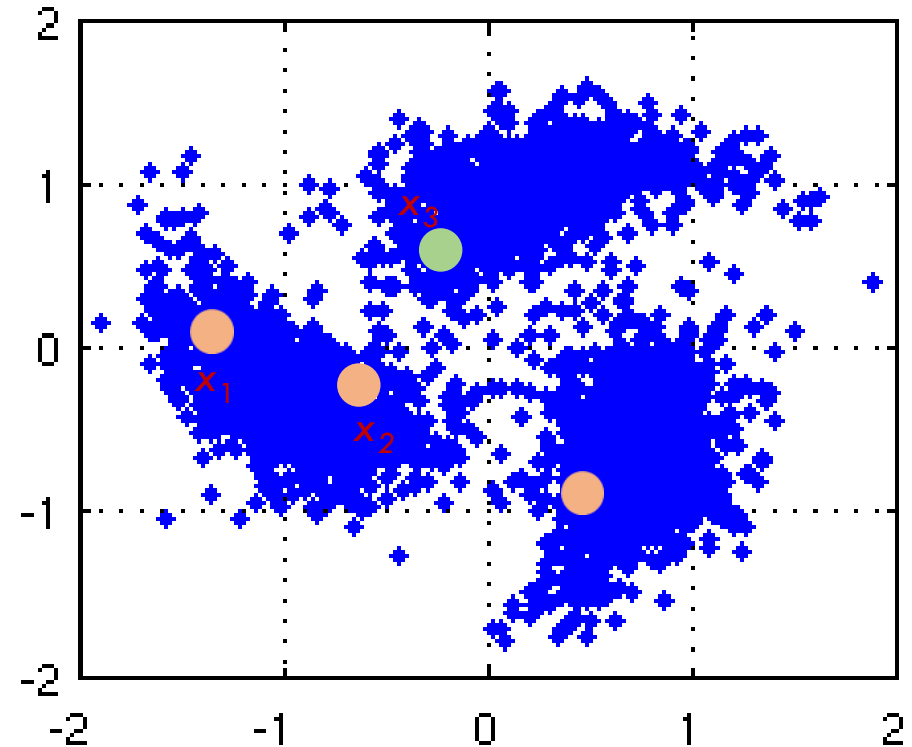


(b) Manifold assumption.

Smoothness Assumption (1)

- Assumption: "similar" instances have the same class label
- More precisely:
 - x is not uniform.
 - If x_1 and x_2 are close in a high-density region, their labels are the same.

connected by a high-density path

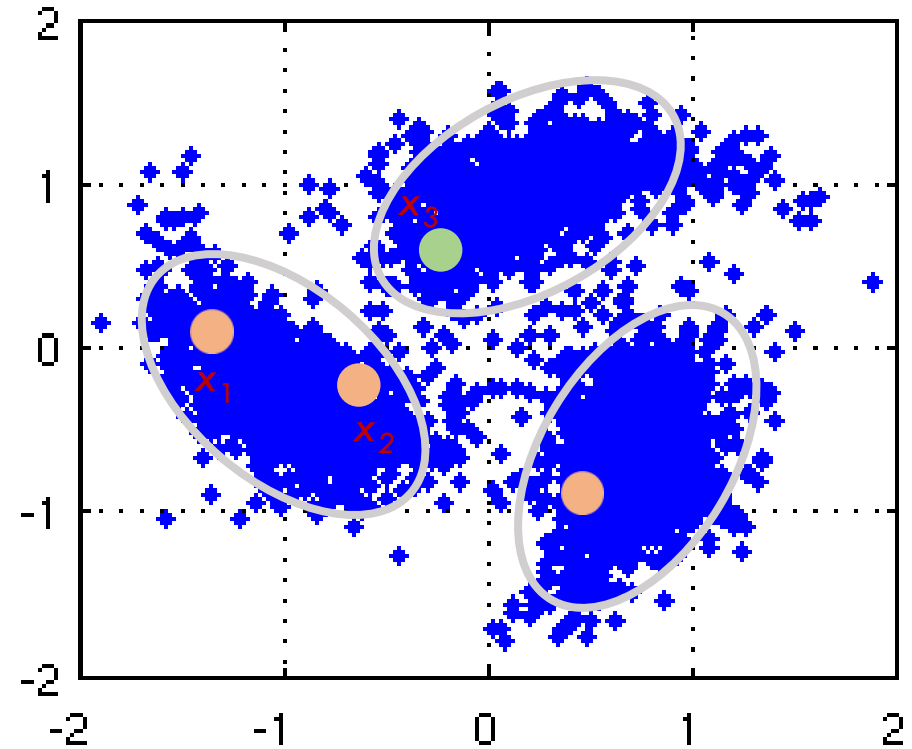


x_1 and x_2 have the same label
 x_2 and x_3 have different labels

Smoothness Assumption (2)

- First cluster the data points
- Then annotate all the data points in a cluster with the same label

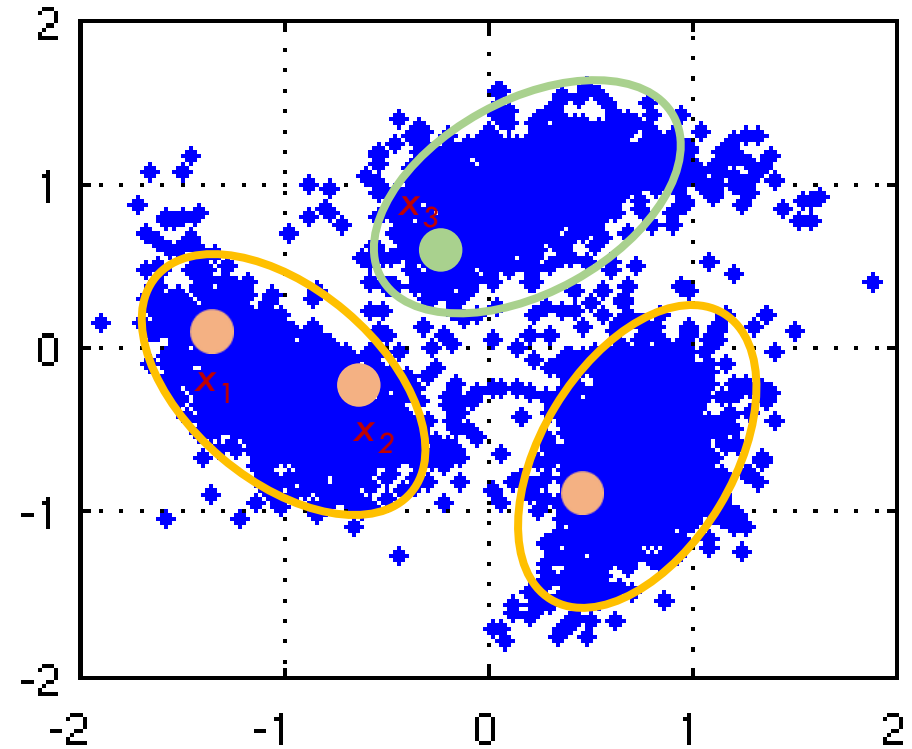
● Class 1
● Class 2



Smoothness Assumption (2)

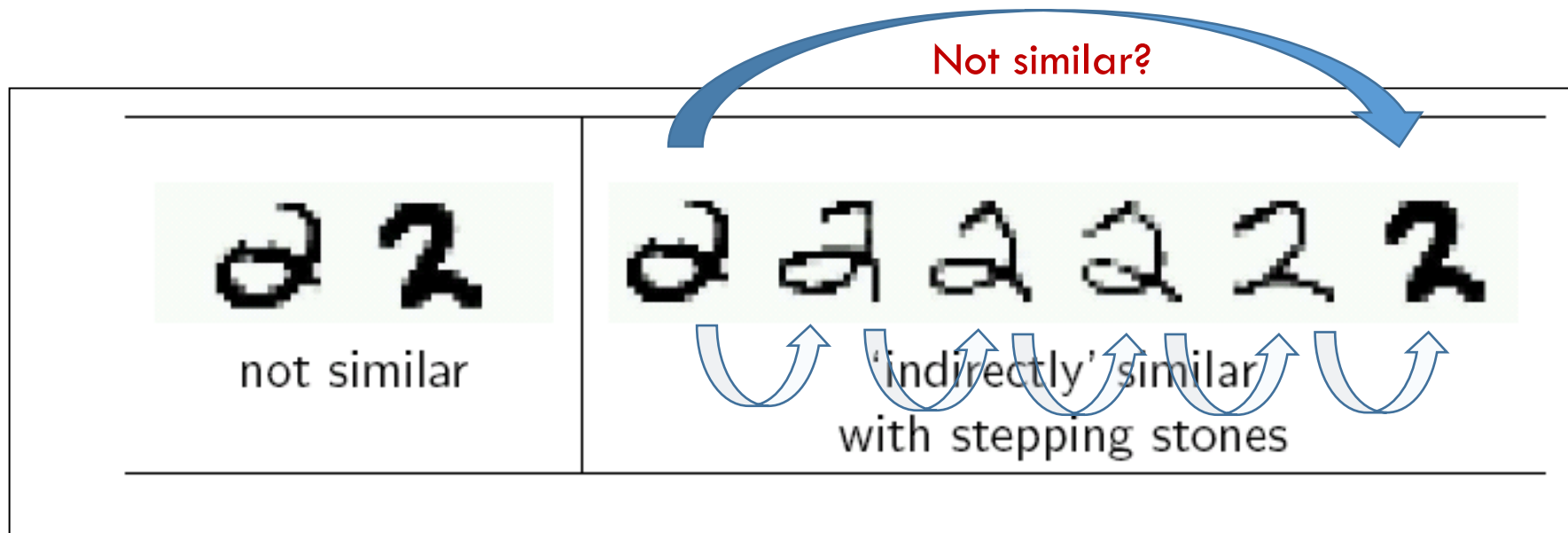
- First cluster the data points
- Then annotate all the data points in a cluster with the same label

● Class 1
● Class 2



Smoothness Assumption (3)

The smoothness assumption can be applied **transitively** to unlabelled data.



Graph-based SSL

- Graph-based SSL algorithms are very attractive:
 - **Convexity:** the majority of algorithms are induced to be convex
 - **Scalability:** able to handle large scale datasets in practice
 - **Effective in practice:** social networks, sensor networks, communication networks, and biological networks are an intrinsic graph structure in practice

Graph-based SSL

- **Transductive algorithms**

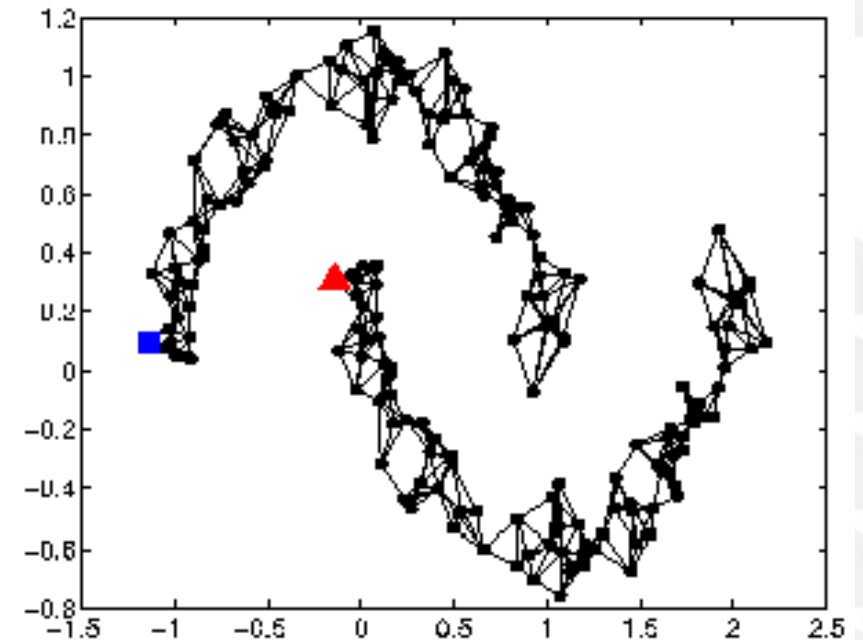
- learn the intrinsic label information of labeled samples, then propagate the label information to the unlabeled samples through graph
- cannot cope with out-of-sample

- **Inductive algorithms**

- can learn a classifier using both labeled and unlabeled samples, which can be used for the classification of unlabeled samples and new out-of-sample

Transductive Graph-based Methods (1)

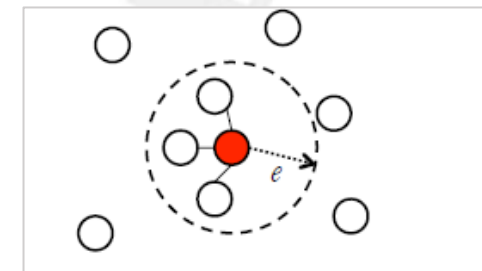
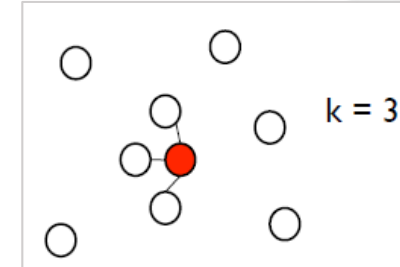
- Represented the data points as a **graph**
- For example, hyperlink of webpages, citation of papers, ...
- Graph representation is natural in many scenarios



Transductive Graph-based Methods (2)

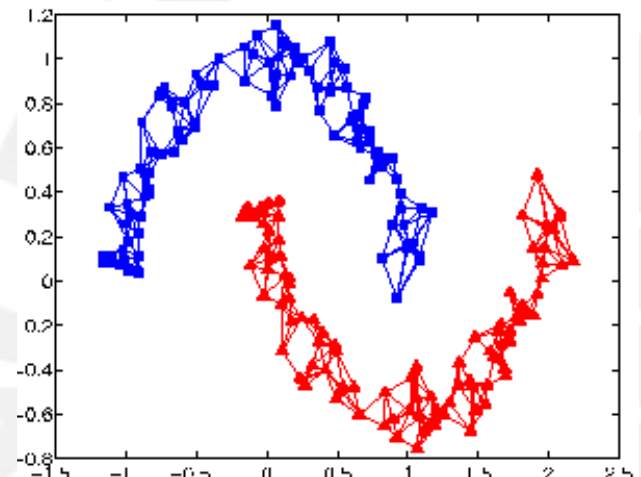
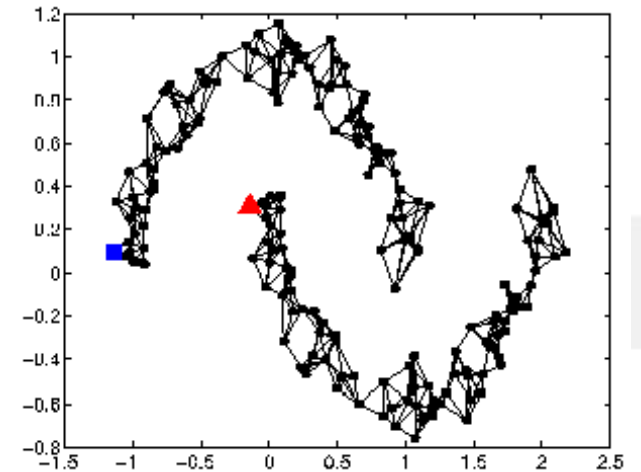
- **Assume** that very similar examples probably have the same label.
- If there are a lot of labeled data, this suggests a Nearest-Neighbor type of algorithm.

- Define the similarity $s(x_i, x_j)$ between x_i and x_j
- Add edge:
 - k Nearest Neighbor
 - e-Neighborhood
- Edge weight is proportional to $s(x_i, x_j)$



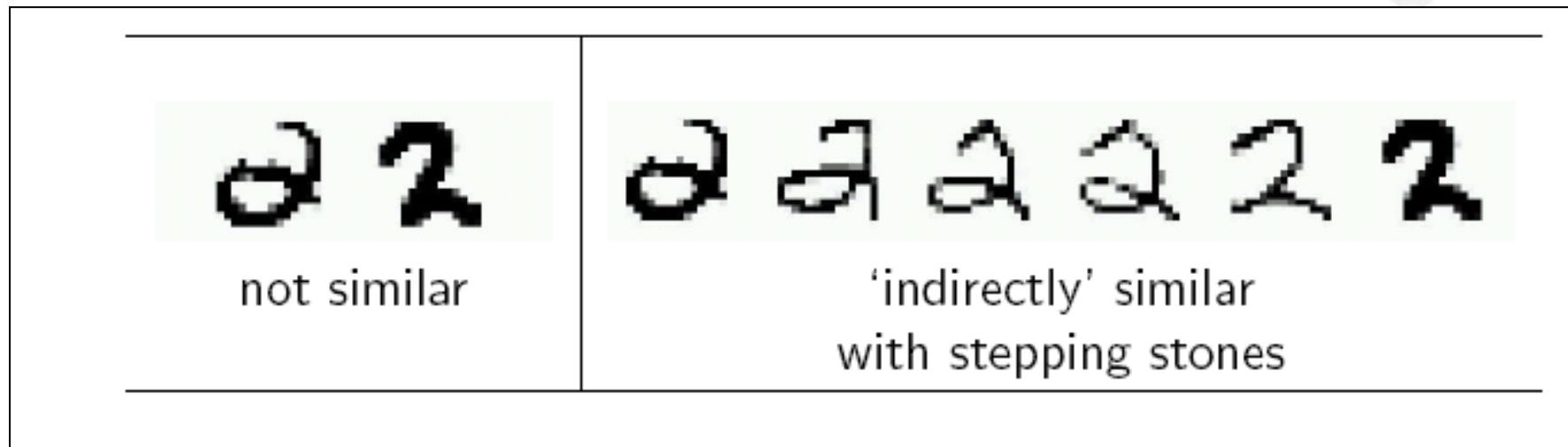
Transductive Graph-based Methods (3)

- Neighbor nodes assumed to be similar with larger edge weights
- Force same class members in L to be close, while maintaining smoothness with respect to the graph for U
- Add in members of U as neighbors based on some similarity
- Iteratively label U (breadth first search)



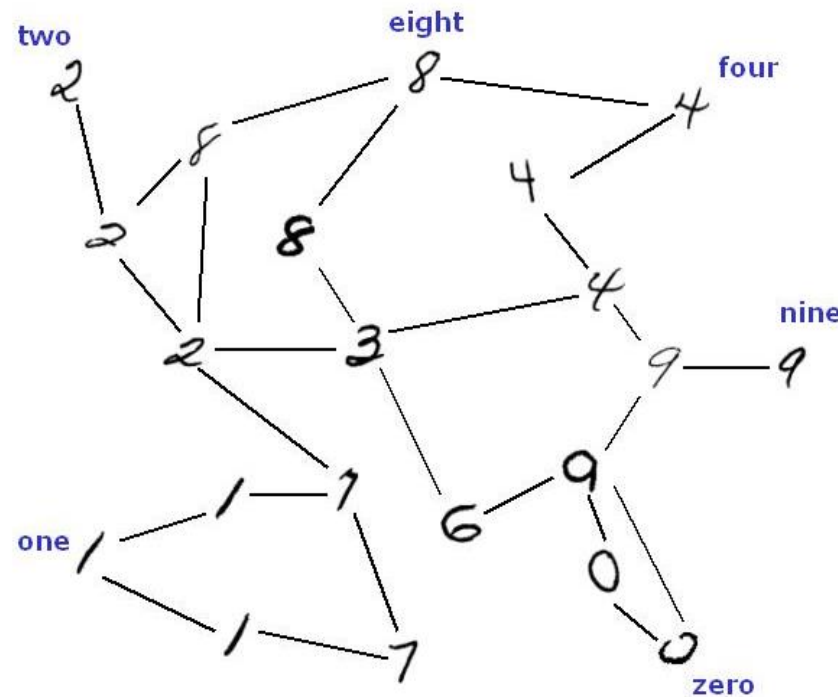
Transductive Graph-based Methods (3)

- **Assume** that very similar examples probably have the same label.
- If there are a lot of labeled data, this suggests a Nearest-Neighbor type of algorithm.
- If there are a lot of **unlabeled** data, perhaps can use them as “stepping stones”, e.g., handwritten digits



Transductive Graph-based Methods (4)

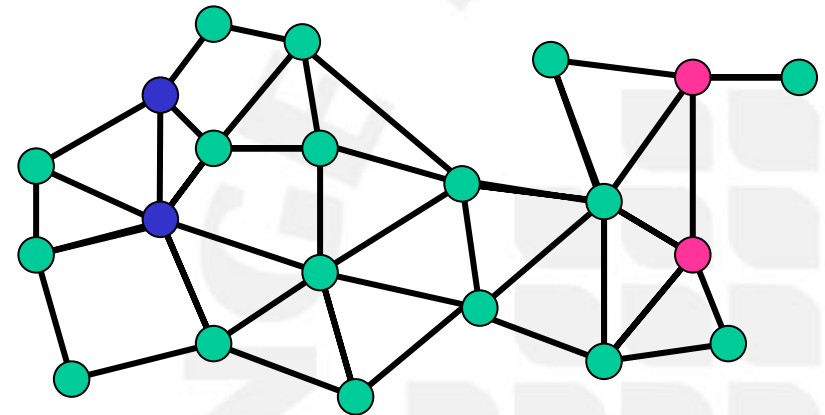
- Idea: construct a graph with edges between very similar examples.
- Unlabeled data can help “glue” the objects of the same class together.



Transductive Graph-based Methods (5)

- **Graph Mincuts**

- Positive samples as sources and negative samples as sinks
- Unlabeled samples are connected to other samples with weights based on similarity
- **Objective:** find a minimum set of edges to remove so that all flows from sources to sinks are blocked



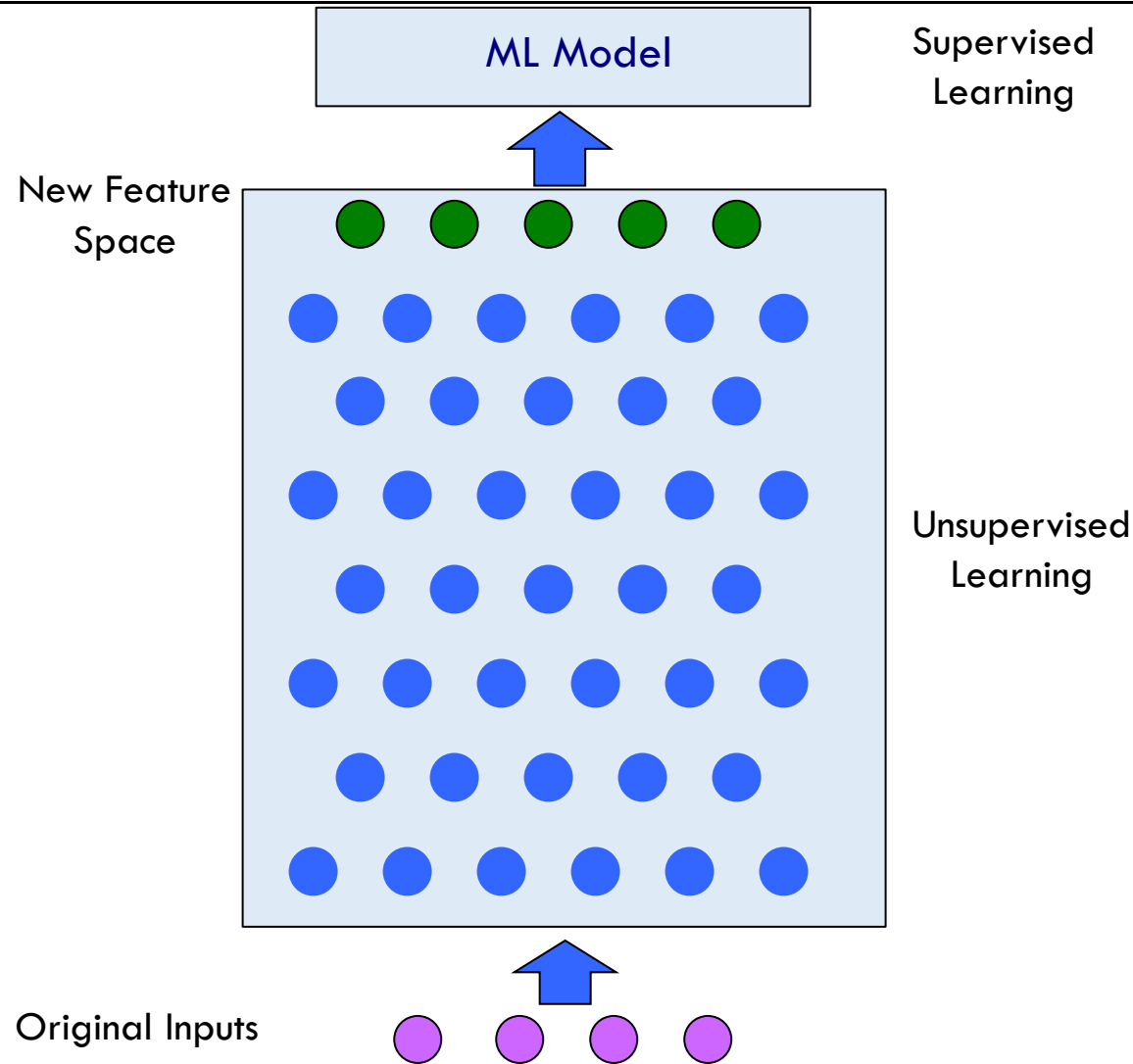
SSL Neural Network-based Methods

- **Supervised neural networks** are trained on large sets of labeled samples
 - creating large labels of samples is very costly and time-consuming task
- **SSL neural network-based methods** combine unsupervised and supervised learning to perform the semi-supervised learning
 - effectively generalizes from small labeled samples to large unlabeled samples
- **Semi-supervised DNN** based on the pseudo-labels of unlabeled samples
 - trained by labeled and unlabeled samples simultaneously in a supervised fashion

DL Semi-supervised Learning Techniques

- Many DL models **learn domain features in unsupervised learning fashion**
 - Deep Belief Nets
 - Sparse Auto-encoders
 - Self-Taught Learning

DL Semi-supervised Learning Techniques



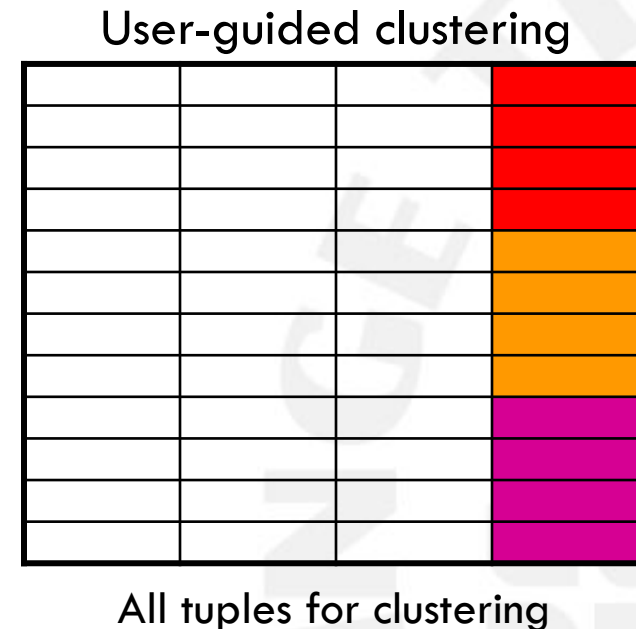
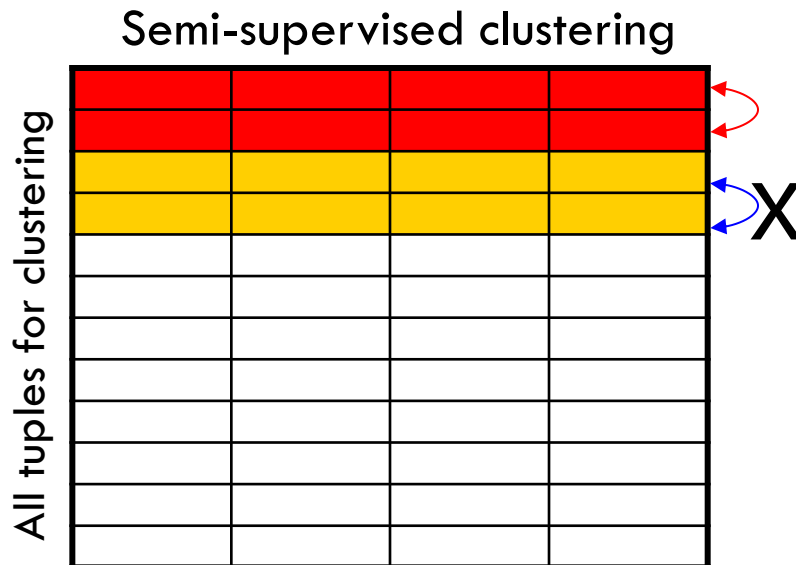
- Many DL models **learn domain features in unsupervised learning fashion**

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Deep Net with Greedy Layer Wise Training

Semi-Supervised Clustering

- **Semi-supervised clustering** - user provides a training set consisting of “similar” (“must-link”) and “dissimilar” (“cannot link”) pairs of objects
- **User-guided clustering** - user specifies an attribute as a hint, and more relevant features are found for clustering



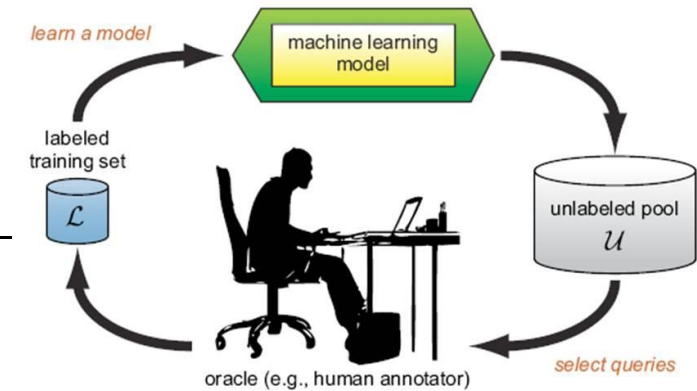
Clustering with constraints

- **Constraints on instances:** specifies how a pair or a set of instances should be grouped in the cluster analysis
 - Must-link vs. cannot link constraints
 - $\text{must-link}(x, y)$: x and y should be grouped into one cluster
 - Constraints can be defined using variables, e.g.,
 - $\text{cannot-link}(x, y)$ if $\text{dist}(x, y) > d$
- **Constraints on clusters:** specifies a requirement on the clusters, e.g.,
 - the min # of objects in a cluster, the max diameter of a cluster, the shape of a cluster
- **Constraints on similarity measurements:** specifies a requirement that the similarity calculation must respect, e.g.,
 - driving on roads, obstacles

Constrained (COP-) k -Means Clustering

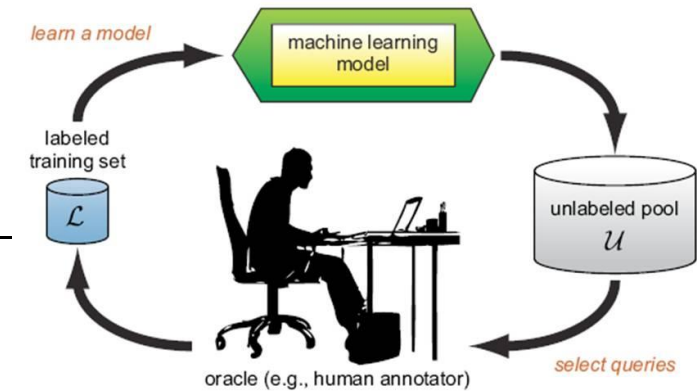
- **Constrained-based clustering** methods respect the constraints in cluster assignments
- **COP- k -means algorithm**
 - Generate super-instances for must-link constraints
 - Compute the transitive closure of the must-link constraints
 - To represent such a subset, replace all those objects in the subset by the mean.
 - The super-instance also carries a weight, which is the number of objects it represents
 - Conduct modified k -means clustering to respect cannot-link constraints
 - Modify the center-assignment process in k -means to a *nearest feasible center assignment*
 - An object is assigned to the nearest center so that the assignment respects all cannot-link constraints

Active Learning (1)



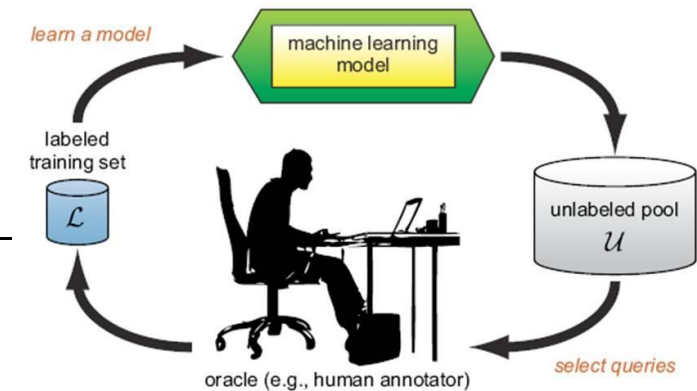
- **Active learner:** query human (oracle) for labels
- Pool-based approach: Uses a pool of unlabeled data
 - L : a small subset of D is labeled, U : a pool of unlabeled data in D
 - Use a query function to carefully select one or more instances from U and request labels from an oracle (a human annotator)
 - The newly labeled samples are added to L , and learn a model
 - **Goal:** Achieve high accuracy using as few labeled data as possible

Active Learning (2)



- Evaluated using **learning curves**: Accuracy as a function of the number of instances queried (# of instances to be queried should be small)
- Selection criteria used to choose the data instances to be queried:
 - **Informativeness** - *reduce uncertainty in a statistical model*
 - Uncertainty sampling
 - Query-by-committee
 - **Representativeness** – *increase the structure representation*
 - Exploit the cluster structure of unlabeled data

Active Learning (2)

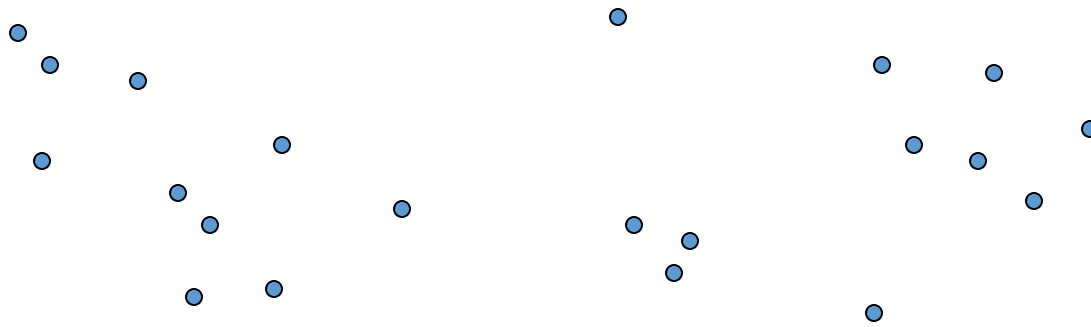


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Often query:

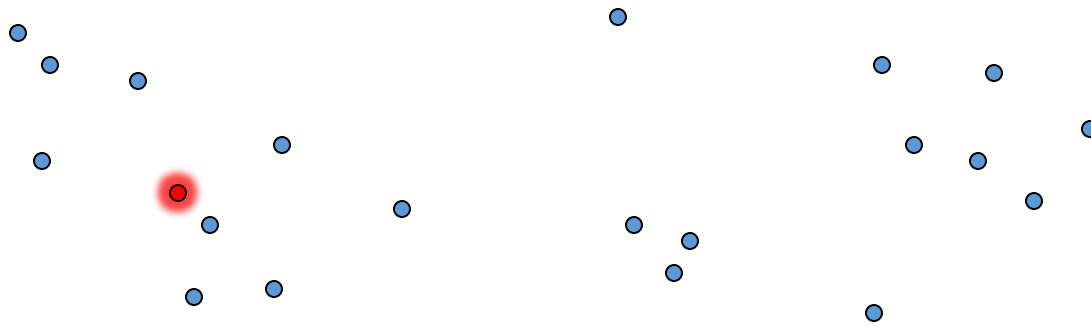
- A low confidence instance (i.e. near a decision boundary)
- An instance which is in a relatively dense neighborhood

Active Clustering



- An instance which is in a relatively dense neighborhood

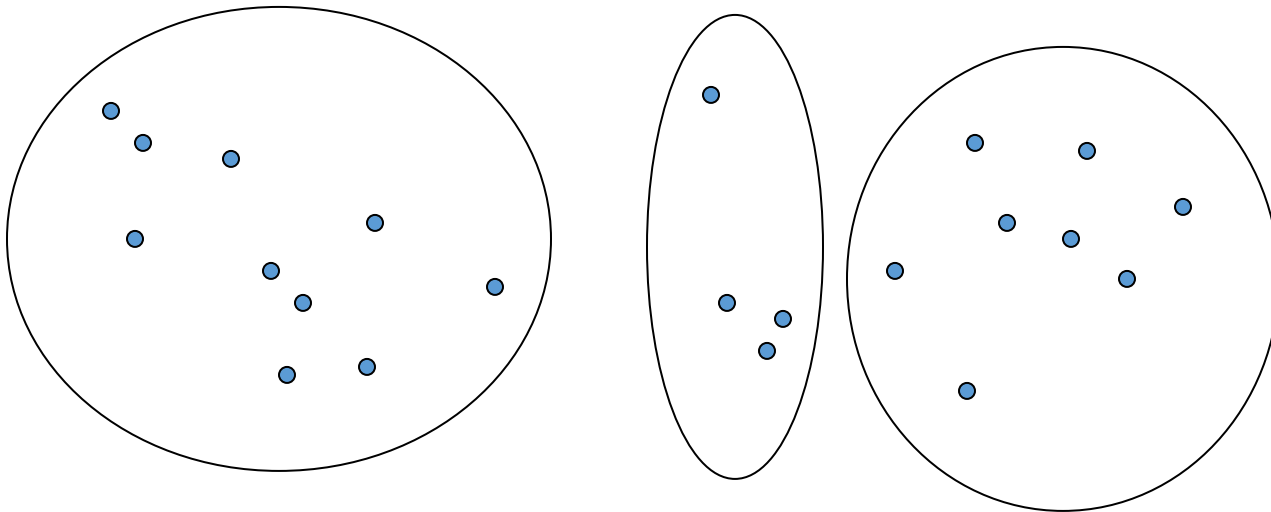
Active Clustering



- An instance which is in a relatively dense neighborhood

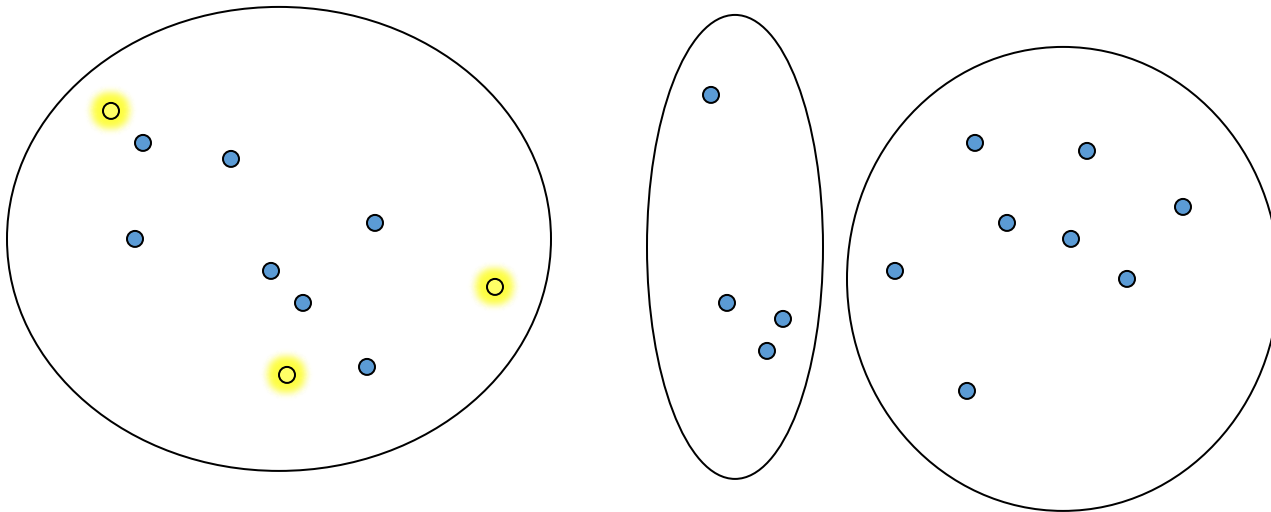
Active Clustering

- A low confidence instance (i.e. near a decision boundary)
 - First do unsupervised clustering
 - Which points to show an expert in order to get feedback on the clustering to allow adjustment?



Active Clustering

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 - First do unsupervised clustering
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Conclusions

- Semi-supervised learning is an area of increasing importance in Machine Learning.
- Automatic methods of collecting data make it more important than ever to develop methods to make use of unlabeled data.
- Several promising algorithms (only discussed a few). Also new theoretical framework to help guide further development.

References

<https://direct.mit.edu/books/book/3824/Semi-Supervised-Learning>

